higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

# MARKING GUIDELINE

## NATIONAL CERTIFICATE (VOCATIONAL)

## MATHEMATICS
(Second Paper)
**NQF LEVEL 2**

**25 FEBRUARY 2019**

**This marking guideline consists of 7 pages.**

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## QUESTION 1

1.1 1.1.1 The median is the middlemost term.

1.1.2 Frequency is the number of times each piece of data is found.

(2 × 2) (4)

1.2 1.2.1 $\text{Mean} = \frac{4069}{24}$ ✓

$= 169{,}542$ ✓ (3)

1.2.2 139; 148; 149; 157; 158; 159; 162; 164; 166; 168; 168; 168; 168; 173; 174; 177; 179; 179; 181; 183; 185; 187; 188; 189

Subtract I mark for every number omitted or written in incorrect order (2)

1.2.3 ✓ ✓

$\text{Median} = \frac{68+68}{2}$

$= 68$ ✓

or alternate

$Q_{2\,position} = \frac{1}{2}(24+1)$

$= 12{,}5$ ✓

$Q_2 = 68 + 0{,}5(68+68)$ ✓

$= 68$ ✓ (3)

1.2.4 Mode of the boys = 168 ✓ (1)

1.3 27 30 34 35 41 | 43 45 46 47 48 | 49 51 52 53 54 | 56 58 63 67 67

1.3 1.3.1 Range = 67 - 27 ✓

$= 40$ ✓ (2)

1.3.2 ✓

$\text{Lower quartile} = \frac{41+43}{2} = 42$ ✓ or

$Q_{1\,position} = \frac{1}{4}(20+1)$

$= 5{,}25$

$Q_1 = 41 + 0{,}25(43-41)$ ✓

$= 41{,}5$ ✓ (2)

1.3.3 ✓

$\text{Upper quartile} = \frac{54+56}{2} = 55$ ✓ or

$Q_{3\,position} = \frac{3}{4}(20+1)$

$= 15{,}75$

$Q_2 = 54 + 0{,}75(56-54)$ ✓

$= 55{,}5$ ✓ (2)

1.3.4 1nterquartile range = 55 - 42 ✓ or 1nterquartile range = 55,5 – 41,5 ✓

$= 13$ ✓ $= 14$ ✓ (2)

1.3.5 $= \frac{40}{100} \times 20$ ✓

Position $= 8$ ✓

$40^{th}$ percentile $= 46$ ✓ (3)

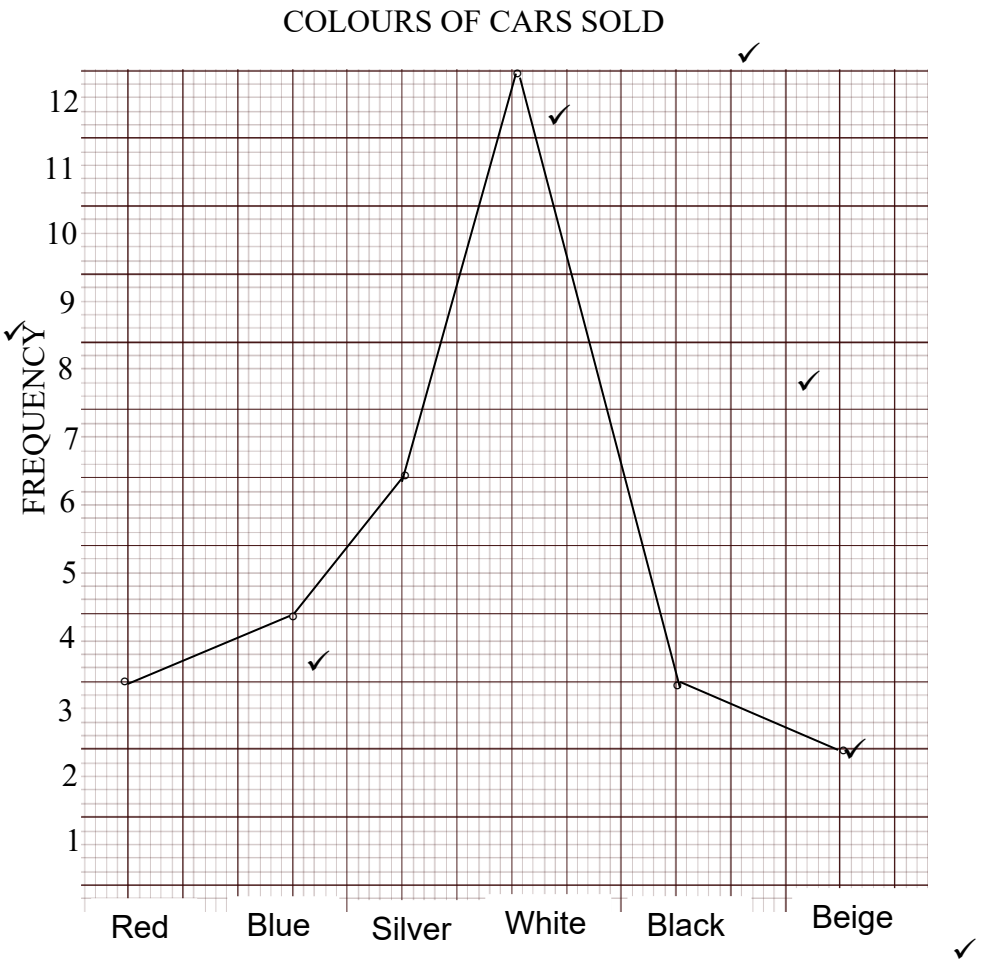
1.4 **ANSWER SHEET**

1.4.1

| **COLOURS OF CARS** | **TALLYy** | **FREQUENCY** |
|---|---|---|
| RED | lll | 3 |
| BLUE | llll | 4 |
| SILVER | ~~llll~~ l | 6 |
| WHITE | ~~llll llll~~ ll | 12 |
| BLACK | lll | 3 |
| BEIGE | ll | 2 |
| | TOTAL | 30 ✓ |

(7)

1. 4. 2




One mark for every two points plotted correctly
One mark for heading
One mark each for labelling horizontal-axis and vertical axis
One mark for joining points

(7)

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1.4.3 White (1)

1.4 1.4.4 Beige (1)
**[40]**

**QUESTION 2**

2.1 2.1.1

$$AB = \sqrt{(x_2-x_1)^2+(y_2-y_1)^2} \qquad CD = \sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$$
$$= \sqrt{(-3--(1))^2+(-4-(-7))^2} \qquad = \sqrt{(2-0)^2+(-5-(-2))^2}$$
$$= \sqrt{4+9} \qquad = \sqrt{4+9}$$
$$= \sqrt{13} \ \checkmark \qquad = \sqrt{13}$$

$$BC = \sqrt{(x_2-x_1)^2+(y_2-y_1)^2} \qquad DA = \sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$$
$$= \sqrt{(-1-2)^2+(-7-(-5))^2} \qquad = \sqrt{(0-(-3))^2+(-2-(-4)^2}$$
$$= \sqrt{9+4} \qquad = \sqrt{9+4}$$
$$= \sqrt{13} \ \checkmark \qquad = \sqrt{13} \ \checkmark$$

Perimeter = AB+BC+CD+DA=14,422 ✓ (5)

2.1.2
$$M\left(\frac{-1+0}{2};\frac{-7-2}{2}\right)$$
$$M\left(-\frac{1}{2};-\frac{9}{2}\right)$$ ✓✓ (2)

2.1.3
$$m_{DC} = \frac{y_2-y_1}{x_2-x_1}$$
$$= \frac{-2-(-5)}{0-2} \ \checkmark$$
$$= \frac{3}{-2}$$
$$= -\frac{3}{2} \ \checkmark$$ (2)

2.1.4
$$midpt_{AC} = \left[\left(\frac{x_1+x_2}{2}\right);\left(\frac{y_1+y_2}{2}\right)\right]$$
$$= \left[\left(\frac{-3+2}{2}\right);\left(\frac{-4+-5}{2}\right)\right]$$
$$= \left(-\frac{1}{2};-\frac{9}{2}\right)$$ ✓✓

Midpoint of AC=M the point has the same co-ordinate ✓ (3)

2.2 2.2.1 2 units to the left and 1unit downwards (2)

2.2.2 $A(2;-1)$
$B(1;-3)$
$C(-1;2)$ (6)

2.3 2.3.1 *Total surface area of triangular prism=(height of prism x perimeter of base)+2(area of base)*

To find the perimeter of the base we need the slant height of the triangle
$=\sqrt{12^2+5^2}$ ✓ for using Pythagoras
$=\sqrt{169}=13$ ✓

*Surface area* = [15 x (12+5+13)]+2($\frac{1}{2}\times 5\times 12$)
$= 450+60$
$= 510\ u^2$

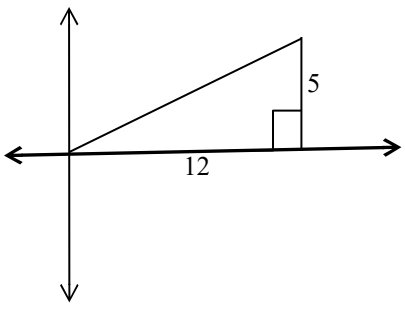
*Or*
*Surface area* = 2 x Area of base(triangle) + (area of horizontal rectangle ) + (area of left side) + (area of right side where you first need to find the length of the slant height of triangle)

$=\left(2\times\frac{1}{2}\times 5\times 12\right)+(5\times 15)+(12\times 15)+(15\times 13)$
$= 510u^2$ (7)

2.3.2 Volume of prism=area of base x height
$V_{triangular\ prism}=\left(\frac{1}{2}\times 5\times 12\right)\times 15$
$= 450u^3$ (3)
**[30]**

**QUESTION 3**

3.1 3.1.1 $r^2=x^2+y^2$
$r^2=(12)^2+(5)^2$ ✓
$r^2=144+25$
$r^2=169$
$r=13$ ✓
$\therefore \sin\theta=\frac{5}{13}$ ✓




(3)

3.1.2
$$= \left(\frac{5}{12}\right)^2 \checkmark - \left(\frac{12}{13}\right)^2 \checkmark$$
$$= \frac{25}{144} - \frac{144}{169}$$
$$= -0,678 \checkmark$$
(3)

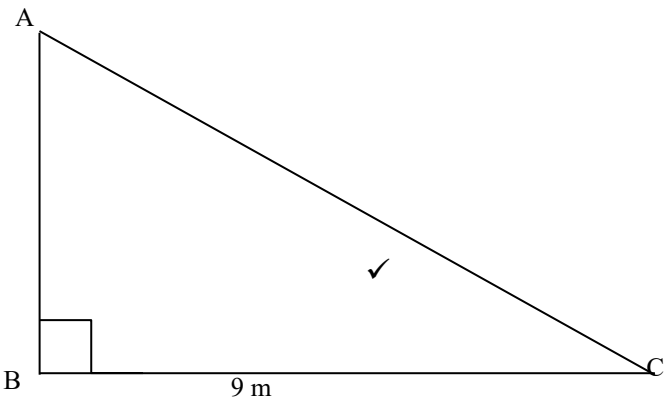
3.2 3.2.1
$$\cos(60^0 - 30^0)$$
$$\cos 30^0 \checkmark$$
$$= 0,866 \checkmark$$
(2)

3.2.2
$$\sin(60^0 - 15^0)$$
$$\sin 45^0 \checkmark$$
$$= 0,707 \; or \; \frac{\sqrt{2}}{2} = \frac{1}{\sqrt{2}} \checkmark$$
(2)

3.3
$$2\cos\theta + 3 = 4$$
$$2\cos\theta = 1 \checkmark$$
$$\cos\theta = \frac{1}{2} \checkmark$$
$$\theta = 60^0 \checkmark$$
(3)

3.4
$$\tan 50^0 = \frac{38}{QR} \checkmark$$
$$QR = \frac{38}{\tan 50^0} \checkmark$$
$$QR = 31,886\,cm \checkmark$$
$$\tan 40^0 = \frac{38}{RS} \checkmark$$
$$RS = \frac{38}{\tan 40^0}$$
$$RS = 45,287\,cm \checkmark$$
$$\therefore QS = 31,886\,cm + 45,287\,cm$$
$$= 77,173\,cm \checkmark$$
(6)

3.5




$\tan 82^0 = \frac{AB}{9\,m}$ ✓

$AB = 9\,m \tan 82^0$ ✓

$AB = 64,038\,m$ ✓ (4)

3.6 3.6.1

$f(x) = b \sin x$

$-1 = b \sin 90^0$ ✓

$b = -1$ ✓

$g(x) = \cos x + p$

$2 = \cos 0^0 + p$ ✓

$2 = 1 + p$

$p = 2 - 1$

$p = 1$ ✓ (4)

3.6.2 $360^0$ ✓ (1)

3.6.3 $Range: y\ \varepsilon\ [-1\,;\,1]$ ✓✓ (2)

**[30]**

**TOTAL: 100**

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